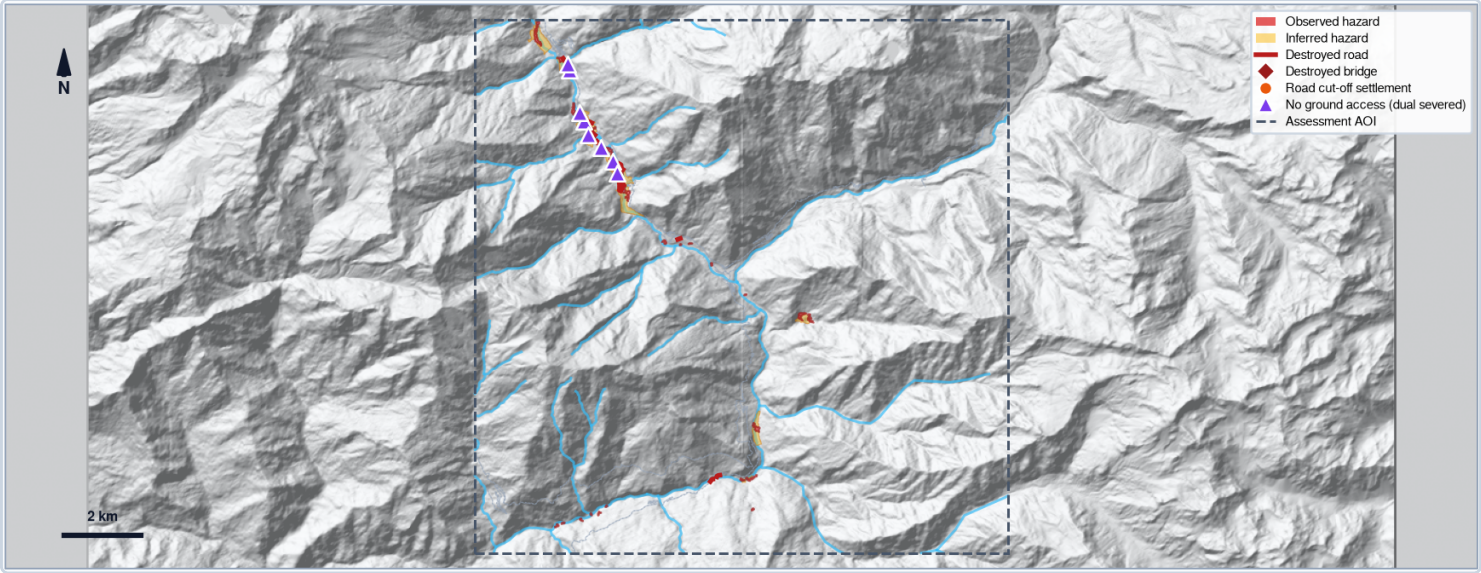
RAPID DISASTER SITUATION REPORT (SITREP)

Educational Prototype

Event Date: **2023-10-04** | Satellite: **2023-10-06 to 2023-10-11** | AOI: **[88.58, 27.53, 88.72, 27.67]**

0.42 km²	47	5	8
OBSERVED HAZARD	DESTROYED BUILDINGS	DESTROYED BRIDGES	CUT-OFF SETTLEMENTS



1. WHERE DID THE FLOOD HIT?

Satellite analysis identifies **0.42 km²** of directly observed hazard extent and **0.98 km²** inferred along tributaries. Affected stretch extends along **Teesta (LachenChu)** from near **Nanga** to near **Nanga** (Sensors: Pre 2023-09-26 to 2023-10-01 / Post 2023-10-06 to 2023-10-11).

2. WHAT WAS DAMAGED?

Asset	Destroyed	Damaged	Cloud-Unv.	Total
Buildings	**47**	20	646	826
Bridges	**5**	0	8	13
Roads (km)	**2.62**	4.74	65.82	75.76

- Major road damage: **residential: 1.77 km** (Total 2.62 km destroyed, 4.74 km possibly damaged, 65.82 km unverified under clouds).
- Major corridor bridges destroyed: North Sikkim Highway (NH310AG), North Sikkim Highway (NH310A).

3. WHO IS CUT OFF?

8 settlements cut off from any town or hospital by road (pessimistic: **10**). **8 settlements** have no ground access (dual vehicle & walking severed).

No Ground Access (Air / Boat Priority)

Settlement	Lat, Lon	Bld
Near Nanga (9.8 km)	27.6297, 88.6172	**47**
Near Nanga (10.6 km)	27.6366, 88.6130	**41**
Near Nanga (11.0 km)	27.6397, 88.6096	**29**
Near Nanga (11.7 km)	27.6456, 88.6074	**17**
Near Nanga (13.0 km)	27.6568, 88.6049	**9**
Near Nanga (13.1 km)	27.6585, 88.6044	**8**
Near Nanga (10.2 km)	27.6329, 88.6162	**7**
Near Nanga (11.4 km)	27.6435, 88.6085	**6**

5 Largest Road Cut-Off Settlements

Settlement	Lat, Lon	Bld
Near Nanga (9.8 km)	27.6297, 88.6172	**47**
Near Nanga (10.6 km)	27.6366, 88.6130	**41**
Near Nanga (11.0 km)	27.6397, 88.6096	**29**
Near Nanga (11.7 km)	27.6456, 88.6074	**17**
Near Nanga (13.0 km)	27.6568, 88.6049	**9**

4. WHAT TO REPAIR FIRST

- No single repair reconnects a settlement on its own.
- **Package 1 (3 elements, 193.7 m) restores vehicle access to 4 settlements (40 buildings).**

Known Data Limitations:
- 646 buildings and 65.82 km roads remain unverified under cloud cover.
- Drainage hazard hydraulically inferred; radar shadow/layover masked with DEM look-angle.
- Asset baseline derived from pre-event OpenStreetMap; census population not included.

Contains modified Copernicus Sentinel data 2023. Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors under ODbL. European Union, Copernicus Emergency Management Service data (EMSR927, EMSR692, EMSR659). Bountos et al., 2024 (Kuro Siwo). Educational prototype, not an operational tool.